MIL-C-43338E
19 December 1979
SUPERSEDING
MIL-C-43338D
22 April 1974

MILITARY SPECIFICATION

CREAM SUBSTITUTE, DRY, NON-DAIRY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers non-dairy cream substitute product for use by the Armed Forces and other authorized Federal agencies, as an item of general use and as a component of operational rations.

2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-636 - Boxes, Shipping, Fiberboard.

MILITARY

MIL-L-35078 - Loads, Unit: Preparation of Nonperishable Subsistence.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-671 - Sanitary Standards for Milk Evaporating and Drying Plants.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Command, Natick, MA 01760, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8940

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(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

LAWS AND REGULATIONS

US DEPARTMENT OF HEALTH, EDUCATION AND WELFARE

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder

(Application for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

AMERICAN PUBLIC HEALTH ASSOCIATION

Standard Methods for the Examination of Dairy Products

(Application for copies should be addressed to the American Public Health Association, 1015 8th Street, N.W., Washington, DC 20036.)

ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, P.O. Box 540, Benjamin Franklin Station, Washington, DC 20044.)

AMERICAN OIL CHEMISTS SOCIETY

Official and Tentative Methods of the American Oil Chemists

(Application for above publications should be addressed to the American Oil Chemists Society, 508 South Sixth Street, Champaign, IL 61820.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington DC 20036.)

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UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Material. The material components shall comply with table I.

TABLE I. Components

Item	Requirement
1	Fat shall be refined hydrogenated vegetable fat and may contain antioxidants, emulsifiers and chelating agents. At time of use, the fat stability shall be not less than 100 hours (active oxygen method) when tested in accordance with 4.4.2.
2	Protein shall be sodium caseinate or other edible proteins.
3	Lactose, dextrose, starch, or corn sirup solids and whey solids may be used singly or in combinations thereof.
4	Emulsifiers, stabilizing salts and free-flowing additives shall comply with 3.5.
5	Artificial flavors and colors shall comply with 3.5.
6	Antioxidants and chelating agents shall comply with 3.5.
7	Salt shall be non-iodized, free-flowing sodium chloride.
8	The water used in the manufacture of the product shall be potable and shall not impart an undesirable odor or flavor to the finished product.

3.2 Processing. The processing shall be in accordance with table II.

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TABLE II. Processing

Item	Requirement
1	Combine all ingredients in a manner suitable to the manufacturing procedure of the contractor.
2	The blend shall be pasteurized by one of the following methods: (a) heating to at least 155°F (68°C) and holding at this temperature for not less than 30 minutes, or (b) any other pasteurization process recognized as equally efficient by the U.S. Public Health Service and approved by the appropriate State or Local Health authority. The product shall be homogenized and spray dried.

3.3 Finished product. The finished product shall comply with table III.

TABLE III. Finished product

Item	Requirement
1	The product shall have a white or light cream color.
2	The product shall be a uniform, granular powder, free-flowing and free from lumps that do not fall apart when pressed between the fingers.
3	There shall be no foreign flavor or odor such as, but not limited to, tallowy, stale, soapy, rancid, bitter or scorched.
4	There shall be no foreign material such as, but not limited to, dirt, insects, insect parts, wood, paint, glass or metal particles.
5	The fat content shall be not less than 33.0 percent.
6	The peroxide value shall not exceed 1.0 milliequivalent per kilogram (meq/kg) of fat.
7	Free fatty acids shall not exceed 0.2 percent.
8	The moisture content shall be not more than 3.0 percent.
9	The standard plate count shall not exceed 20,000 per gram.
10	The coliform count shall not exceed 10 per gram.

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TABLE III. Finished product

Item	Requirement
11	When examined for performance in hot coffee, the cream substitute shall sink and disperse within 15 seconds. There shall be no evidence of curdling (feathering) and no particles which rise again and remain floating on the surface of the coffee, except for a trace of melted fat. After observing the above, stir the coffee for 15 seconds. Observe the whitening power which shall be equivalent to the color rendered by addition of homogenized milk to coffee prepared in the same manner (see footnote 1/). There shall be no sediment on the bottom of the vessel after one minute of standing following the stirring.

1/ To 8 ounces (227 ml) of potable water at 175°F to 180°F (79°C - 82°C) contained in a clean transparent glass vessel, add 2.5 grams of instant coffee. Add 4 grams of dry cream substitute and observe for requirements. To a **second 8 ounce (227 ml) portion of prepared coffee add 30 milliliters of homogenized milk.**

3.4 Plant qualification. The product shall originate, be processed, marked and handled in plants which meet the requirements of MIL-STD-671.

3.5 All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Plant qualification conditions. The product furnished under this document shall be unacceptable if not produced and stored in plants currently meeting the qualification conditions of 3.4.

4.3 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

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4.3.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.3.1.1 Examination of ingredients. Examinations pertinent to ingredients referenced in 3.1 shall be performed in accordance with 4.3.1.1.1 and 4.3.1.1.2. Use of a nonconforming ingredient, as indicated by examination, shall be cause for rejection of the involved finished product.

4.3.1.1.1 Identity. Conformance of each ingredient to identity requirements (kind, food grade quality, FDA approval, etc., as applicable) shall be ascertained by examination of labels, invoices, certificates or other valid documents.

4.3.1.1.2 Condition. Each ingredient shall be examined organoleptically, as necessary, to determine conformance to condition requirements referenced in 3.1, as applicable. The sample unit for examination shall be approximately 1 pound (.45 kg) derived from a primary container. Five containers shall be sampled from each lot of ingredient. In the event a lot contains less than five containers, each container in the lot shall be sampled.

4.3.1.2 Testing of components. In addition to quality assurance provisions of referenced documents, components shall be tested as shown in table IV. Results shall be applicable to the lot average. Nonconformance to one or more test requirements shall be cause for rejection of the lot.

TABLE IV. Component test

Component	Sample unit	Lot size expressed in	Inspection level	Characteristics	Results reported	Test ref & reqmt para
Hydrogenated vegetable fat	1 pound composite	Pounds	S-1	Stability	Pass or fail 1/	4.4.2 3.1

1/ If failure, report actual time in hours.

4.3.1.2.1 Envelope material certification. Conformance to thickness of envelope laminae foil and polyethylene components and basis weight of paper component requirements shall be determined by examination of pertinent labels, invoices, or other valid documents.

4.3.2 In-process examination. Examination shall be made during processing to determine compliance with the operating procedures referenced in 3.2. Temperature(s) and holding time(s) records shall be maintained. Noncompliance with one or more requirements as determined by actual examination or as reflected by examination of records shall be cause for rejection of the involved lot of finished product.

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* 4.3.3 End item inspection. Classification of defects found during examination shall be in accordance with tables VI through X and 4.3.3.1. The examination criteria for the above tables shall be as shown in table V. The acceptable quality levels (AQLs) shall be expressed in terms of defects per hundred units for tables VI and VIII and percent defective for tables VII, IX and X.

* TABLE V. End item examination criteria

Table	Inspection level	Sample unit	Lot size expressed in	AQLs Major A	Major B	Minor
VI	I	One envelope	Number of envelopes	0.25	1.5	4.5
VII	S-3	Contents of one envelope	Number of envelopes	---	---	1.5
VIII	S-3	Contents of one envelope	Number of envelopes	1/	0.65	4.5
IX	S-3	One filled and sealed envelope	Envelopes	---	1.0	---
X	S-3	One tray or carton	Number of trays or cartons	---	1.5	1.0

1/ See footnote 1/ to table VIII.

TABLE VI. External examination of envelopes

Category			Defect
Major A	Major B	Minor	
101			Tear or hole
	151		Not 3 gram or 4 gram, as applicable
	152		Envelopes exceed maximum length and width (3-1/2 by 2-3/4 inches) (89 mm by 70 mm) (level A preservation only)
	153		Thickness of filled and sealed envelopes exceeds 5/32 inch (4 mm) (level A preservation only) 1/
		201	Outside of envelope not a dull brown or light yellow (level A preservation only)
	154		Crushed or misshapen
	155		Objectionable odor

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* TABLE VI. External examination of envelopes (cont'd)

Category			Defect
Major A	Major B	Minor	
		202	Evidence of delamination.
	156		Not clean.
	157		Nomenclature missing, incorrect or illegible.
		203	Required labeling information other than nomenclature missing, incorrect or illegible.
	158		Envelope not heat sealed on all four edges with minimum 1/8 inch (3 mm) wide seals.

1/ Envelopes to be laid on a horizontal surface and powder evenly distributed throughout inner surface prior to measurement.

* TABLE VII. Examination for net weight

Category	Defect
Minor	
201	Net weight of individual envelope less than 3.7 grams. Report results to the nearest 0.1 gram (level A preservation only). (For envelopes requiring 4 grams). 1/
202	Net weight of individual envelope less than 2.7 grams. Report results to the nearest 0.1 gram. (level C preservation for envelopes requiring 3 grams) 1/

1/ The lot shall be rejected if the sample data indicate a lot average net weight less than the specified net weight per envelope.

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TABLE VIII. Examination for product characteristics

Category			Defect
Major A	Major B	Minor	
101			Presence of foreign material, e.g., rust, dirt, paint, glass, metal, wood, insect(s), insect part(s) or other extraneous material. 1/
	151		Presence of any objectionable flavor or odor, e.g., rancid, stale, oxidized, bitter, soapy, tallowy, scorched, etc.
	152		Lumps that do not fall apart under light pressure between the fingers.
	153		Not a free flowing granular powder.
		201	Not white or light cream color.

1/ Evidence of this defect shall be cause for rejection of the lot.

TABLE IX. Examination for leakage (level A) preservation 1/

Category	Defect
Major B	
151	Leakage.

1/ See 4.4.1 for examination procedure.

* TABLE X. Examination of trays or cartons

Category		Defect
Major B	Minor	
151		National Stock Number, item description , contract number or date of pack markings, missing, incorrect or illegible.
	201	Less than 50 or 100 envelopes per tray or carton. 1/
	202	Arrangement not as specified.
	203	Depth of tray not equal to full height dimension of envelope.

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* TABLE X. Examination of trays or cartons (cont'd)

Category	Defect
Minor	
204	Cartons not securely closed.
205	Other required marking, missing, incorrect, or illegible (cartons only).
206	Not closed in the manner specified.
207	Carton insecurely closed or damaged.

1/ The lot shall be rejected if sample data indicate a lot average count less than the required amount.

4.3.3.1 Hot coffee performance examination. The finished product shall be examined to determine conformance to the performance requirements of item 11 of table III. The lot size shall be the number of pounds (Kg) in a single day's production. The sample unit shall be a 10 gram sample randomly selected from the lot. The sample for inspection shall be a 4 gram composite drawn from the number of sample units called for by inspection level S-3. Nonconformance to one or more of the requirements shall be cause for the rejection of the lot.

4.3.4 End item testing. The finished product shall be tested for requirements specified in 3.3. Procedures for testing for chemical and bacteriological requirements shall be in accordance with 4.4.2 and 4.4.3, respectively. Requirements for all tests, except bacteriological requirements shall be lot average requirements. The sample unit for testing shall be one envelope. Lot size shall be expressed in terms of the sample unit. Sample size shall be the number of envelopes indicated by inspection level S-4. Sample units of the product shall be composited, and approximately an 8-ounce (227 g) composite of the finished product shall be prepared for testing. Samples for lot average requirements shall be composited by the laboratory. Five envelopes shall be withdrawn and tested individually for bacteriological requirements. Test results shall be reported as follows:

Fat, moisture to the nearest 0.1 percent.

Peroxide value to the nearest 0.1 meq/kg of fat.

Free fatty acids to the nearest 0.1 percent.

Standard plate and coliform count in accordance with Standard Methods for the Examination of Dairy Products.

Nonconformance to one or more test requirements shall be cause for rejection of the lot.

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4.3.5 Packaging inspection.

4.3.5.1 Levels A and B packing. Examination of the filled and closed shipping containers shall be in accordance with the examination criteria contained in the appendix to PPP-B-636 with the following defects added to the table: Major: National stock number, item description, contract number or date of pack markings missing, incorrect or illegible. Minor: (1) Weight contents of filled shipping container exceeds 45 pounds (20 Kg). (2) Tray or carton missing. (3) Other required markings missing, incorrect or illegible. (4) When applicable, reinforcement with nonmetallic strapping or tape is not used.

4.3.5.2 For level C packing (for shipment to ration assembly). Examination shall be in accordance with table XI. The sample unit shall be one filled and marked shipping container. The sample size for examination shall be the number of shipping containers indicated by the inspection level S-3. The AQLs, expressed as defects per hundred units, shall be 1.5 for major defects and 6.5 for minor defects.

* TABLE XI. Examination of level C shipping containers

Category		Defect
Major B	Minor	For shipment to ration assembly
151		Item description National Stock Number, contract number or date of pack markings, missing, incorrect or illegible.
152		Not packed as specified.
	201	Count more than 2 percent under marked count. 1/ 2/
	202	Weight of product of filled shipping container exceeds 45 pounds (20 Kg).
	203	Other required markings, missing, incorrect or illegible.
	204	Not closed as specified.

1/ The lot shall be rejected if the sample data indicate a lot average count less than the specified count.

2/ If packed in cartons, examine one carton per shipping container.

4.3.5.3 Unit loads. Unitized loads shall be examined in accordance with the criteria of MIL-L-35078 and the applicable specification sheet for the type and class of load specified.

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4.4 Methods of inspection.

4.4.1 Leakage test. The filled and sealed envelope shall be examined for leakage by submerging in water, contained in a desiccator or other suitable container and maintaining a vacuum of 10 inches of mercury (33.76 Kpa) (atmospheric pressure 29.9 inches) (101.2 Kpa) for at least 30 seconds. A leak is indicated by a steady progression of bubbles. Isolated bubbles caused by entrapped air are not considered as leaks.

4.4.2 Chemical analyses. Chemical analyses shall be made in accordance with the following methods:

Test	Source	Method
Fat	1/	Fat in dried milk (method B)
Fat stability	2/	Cd 12-57
Peroxide value	3/	3/
Free fatty acids	3/	3/
Moisture	1/	Moisture

1/ Association of Official Analytical Chemists, Official Methods of Analysis, Chapter: Dairy Products; Section: Dried Milk and Malted Milk.

2/ American Oil Chemists Society, Official and Tentative Methods.

3/ See 4.4.2.1.

4.4.2.1 Determination of peroxide value and free fatty acids. Test for determination of peroxide value and free fatty acids shall be as follows:

a. Preparation of chloroform extract

1. Blend 50 grams (g) of product, 100 milliliters (ml) of chloroform and 15 g of anhydrous sodium sulfate for 3 minutes in a blender or similar device.

2. Filter the blend through No. 12 Whatman filter paper into a 250 ml glass stoppered flask.

3. Rinse blender four times with 50 ml of chloroform and pour through the filter. Between 150 and 200 ml of filtrate is necessary.

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a. Preparation of chloroform extract (cont'd)

4. If filtrate is not clear, treat with additional sodium sulfate and refilter. Repeat until filtrate is clear and bright.

5. Determine the weight of fat per 25 ml aliquot by pipetting an aliquot into a tared fat extraction dish, evaporating the chloroform in a draft oven 100°F (38°C), cooling and weighing.

b. Determination of free fatty acids

1. Pipette a 25 ml aliquot of the chloroform extract into a 250 ml glass-stoppered flask.

2. Add 50 ml of chloroform and titrate the mixture with 0.01N alcoholic KOH using 1 ml of one percent phenolphthalein as indicator.

3. Conduct a blank determination.

4. Calculate free fatty acids as follows:

$$\% \text{ free fatty acids (as oleic)} = \frac{\text{ml of KOH 0.01N X 0.282}}{\text{wt. of fat in 25 ml aliquot (g)}} \text{ X } 100$$

c. Determination of peroxide values

1. Pipette 25 ml of chloroform extract into a 250 ml glass stoppered flask.

2. Add 30 ml of acetic acid and mix well.

3. Add 0.5 ml of saturated potassium iodide solution. Allow the reaction to proceed for exactly one minute with occasional swirling, then add 30 ml of distilled water to stop the reaction.

4. Titrate with 0.01N sodium thiosulfate until the yellow color has almost faded.

5. Add 0.5 ml of starch indicator solution and, with vigorous agitation, continue the titration until the blue color of the starch-iodine complex has faded leaving a clear, colorless solution.

6. Conduct a blank determination.

7. Calculate the peroxide value as follows:

$$\text{Peroxide value as milliequivalents of peroxide per Kg of fat} = \frac{\text{ml of titrating solution X normality X 1000}}{\text{wt. of fat in 25 ml aliquot}}$$

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4.4.3 Bacteriological analyses. Bacteriological analyses shall be made in accordance with the following methods published in Standard Methods for the Examination of Dairy Products.

Test	Chapter	Method
Standard plate count	1/	Agar plate method
Coliform count	1/	Coliform group in dry milk

1/ Chapter: Microbiological Methods for Concentrated Milk and Dry Milk.

5. PACKAGING

* 5.1 Preservation (level A or C). Cream substitute shall be preserved in accordance with level A or level C, as specified (see 6.2).

* 5.1.1 Level A. Not less than 4 grams of the cream substitute shall be filled into an envelope having outside dimensions of not more than 3-1/2 by 2-3/4 inches (89 mm x 70 mm). A minus tolerance of 0.3 g is allowed in any one envelope providing that the average net weight of the lot is not less than 4 g. The envelope shall be made from a heat-sealable laminated material consisting of 25 to 30 pounds (42 to 50 g/m^2) basis weight (24 by 36-500) kraft or sulfite paper - 0.00035 to 0.0005 inch (0.0089 mm to 0.0127 mm) thick aluminum foil - 0.001 to 0.002 inch (0.025 mm to 0.051 mm) thick polyethylene. Adhesives used in laminating shall be permanently plastic materials other than wax and the material shall show no evidence of delamination after heat sealing. The total thickness of the filled, sealed unit pack shall not exceed 5/32 inch (4 mm). All four edges of the envelope shall be heat-sealed with seals not less than 1/8 inch (3 mm) wide. The sealed unit pack shall show no leakage when examined in accordance with 4.4.1. The exterior color of the unit pack shall be as specified in 5.4.1. There shall be no crushed, mishapened or unclean envelopes.

* 5.1.2 Level C. Cream substitute shall be preserved in individual serving size envelopes as specified (see 6.2). Three grams of the product, as specified (see 6.2) shall be unit packed in a foil laminated envelope normally used commercially for the product. A minus tolerance of 0.3 g is permitted in any envelope provided that the average net weight of the lot is not less than 3 g.

5.2 Packing. Cream substitute shall be packed in accordance with level A, B, or C, as specified (see 6.2).

5.2.1 Level A packing. Fifty or one hundred envelopes of cream substitute shall be regularly arranged, on edge, in a paperboard tray, the depth of which shall be equal to the full height dimension of the envelope. Alternatively,

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50 or 100 envelopes shall be packed lying flat in a folding carton. The carton shall be securely closed. The number of envelopes per tray or carton shall be the same for all trays and cartons. Not more than 45 pounds (20 kg) of the product in trays or cartons shall be packed in a snug-fitting fiberboard box constructed, closed and reinforced in accordance with style RSC, V2s of PPP-B-636.

5.2.2 Level B packing. The envelopes in trays or cartons, in the quantity and arrangement and within weight limitation specified in 5.2.1, shall be packed in a snug-fitting fiberboard box, constructed, closed and reinforced in accordance with style RSC, V3c, V3s or V4s of PPP-B-636.

5.2.2.1 Reinforced shipping containers. When specified (see 6.2 and 6.3), the shipping container specified in 5.2.2 shall be reinforced with nonmetallic strapping or pressure-sensitive adhesive, filament reinforced tape in accordance with the appendix of PPP-B-636, except that two reinforcing bands may be used, one lengthwise and one girthwise.

5.2.3 Level C packing. The envelopes, packed in trays or cartons as specified in 5.2.1, shall be packed in a manner to ensure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall comply with the Uniform Freight Classification or National Motor Freight Classification, as applicable, except fiberboard containers shall be closed in accordance with method II as specified in the appendix of PPP-B-636.

5.2.3.1 For shipment to ration assembly. One or two thousand envelopes, as specified in 5.1.1, shall be packed into a shipping container complying with 5.2.3. Alternatively, 1000 or 2000 envelopes may be packed into a paperboard carton and not more than 45 pounds (20 kg) of the product, so packed, shall be filled into a shipping container complying with 5.2.3.

5.3 Unit loads. When specified (see 6.2), the cream substitute, packed in shipping containers in accordance with 5.2, shall be prepared in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2).

5.4 Labeling and marking.

5.4.1 Envelopes for rations. The outside of the completed envelope shall be a dull brown or light yellow. The following information shall be printed in black on the envelope and at random, provided that the complete information appears at least once on envelope:

CREAM SUBSTITUTE, DRY, NON-DAIRY
(In letters larger than any others used for the label).

For coffee or tea.

4 grams net.

Dissolve contents in 1/3 canteen cup (8 oz.) of beverage.

Name and address of manufacturer.

Any other information required by the Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

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5.4.2 Unit packs in accordance with 5.1.2. Any commercial labeling that complies with the Federal Food, Drug and Cosmetic Act and regulations promulgated thereunder, will be acceptable.

5.4.3 Shipping containers and cartons.

5.4.3.1 Marking of shipping containers. Shipping containers and cartons shall be marked in accordance with MIL-STD-129.

5.4.4 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078.

6. NOTES

6.1 Intended use of product and technical information. The product is intended to be used for creaming of coffee or tea by direct addition of the dry product to the hot liquids. No other preparation is necessary. It is a substitute product for fresh cream or other dairy products used for creaming hot liquid drinks. The product offers advantages in logistics in that it is a highly concentrated product. Product has excellent shelf life and is acceptable even after one year at a constant temperature of 100°F (38°C). Storage in a cool dry location, however, will aid in extending its shelf life.

* 6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Levels of preservation and packing required (see 5.1 and 5.2).
(c) When cream substitute is to be packaged in 3-gram or 4-gram individual serving size envelopes (see 5.1.1 or 5.1.2).
(d) When nonmetallic reinforcement is required for level B packing (see 5.2.2.1).
(e) Type and class of unit load when unit laoding is specified (see 5.3).

6.3 Selection of nonmetallic reinforcement. The nonmetallic reinforcement specified in 5.2.2.1 should be selected for shipping containers that are destined for transfer at sea or other specific overseas operations.

6.4 Appropriate level of pack. Based on the conditions known or expected to be encountered during shipment, handling, and storage of the specific item being procured, the contracting officer should select the appropriate level of pack in accordance with the criteria established in AR 700-15, NAVSUPINST 4030.28, AFR 71-6, MCO 4030.133A or DLAR 4145.7, as applicable.

6.5 Metric equivalents. Metric equivalents, indicated in parentheses throughout this document, are based on practices, conversion factors, and symbols specified in ASTM E 380 Standard for Metric Practice, and are for information only. In each instance, the value stated in US customary units shall be controlling.

6.6 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Custodians:

Army - GL
Navy - SA
Air Force - 50

Review activities:

Army - MD
Navy - MC, MS
DP - SS

Preparing activity:

Army - GL

Project No. 8940-0308

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